

EMCH 201: Introduction to Application of Numerical Methods

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Assignment 3

Assignment due date: **11:59 PM, 11th October , 2021**

Point Possible: 100 (30+30+40)

Submission requirements:

- Detailed problem-solving process needs to be provided for full points
- Upload readable electronic copy to blackboard in a single file
- Late submission will not be graded

1. Given the matrix $[A]$ and $[B]$, $[A] = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 0 \\ 4 & 2 & 9 \end{bmatrix}$ $[B] = \begin{bmatrix} 3 & 2 & 0 \\ 1 & 1 & 2 \\ 3 & 5 & 1 \end{bmatrix}$, please calculate:

- $[A]^T$
- $[A]+[B]$
- $[A]-[B]$
- $[A]\cdot[B]$
- $[B]\cdot[A]$

2. Use Gauss Elimination method to solve

$$\begin{aligned} 7x_1 + 2x_2 - 3x_3 &= -12 \\ 2x_1 + 5x_2 - 3x_3 &= -10 \\ x_1 - x_2 - 6x_3 &= -26 \end{aligned}$$

- Put the equation in the matrix form
- Show the forward elimination steps
- Show the backward substitution steps

3. Solve the following system of equations by LU decomposition. Show all the steps of your calculation.

$$\begin{aligned} 15x_1 + 7x_2 - 4x_3 &= 51 \\ 4x_1 - 4x_2 + 9x_3 &= 62 \\ 12x_1 - x_2 + 3x_3 &= 8 \end{aligned}$$

- The decomposition/elimination step
- The forward substitution step
- The backward substitution step